

Entropy-Free Compression via Unconstrained Vector Quantization: Formal Proofs and Rate Analysis

Walker Kirkpatrick

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Abstract

This document provides formal proofs for the theoretical foundations of entropy-coding-free compression via unconstrained vector quantization (VQ), extending the EF-LIC framework of Cao et al. (ICML 2026). We prove three main results: **(1)** the index distribution of an unconstrained VQ converges to the maximum-entropy (uniform) distribution as the codebook size grows; **(2)** the rate penalty of entropy-free versus entropy-coded coding is bounded by the entropy gap and vanishes asymptotically; **(3)** entropy-free coding is optimal if and only if the index distribution is uniform, indices are independent, and context is information-irrelevant. A corollary characterizes the $3\times$ encoding and $5\times$ decoding speedup achieved by removing sequential entropy coding.

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1 Setup and Definitions

Definition 1.1 (Unconstrained Vector Quantizer). An *unconstrained vector quantizer* Q_K with codebook $\mathcal{C}_K = \{c_1, \dots, c_K\} \subset \mathbb{R}^d$ maps a source vector $X \in \mathbb{R}^d$ to the index of its nearest codeword:

$$Q_K(X) = \arg \min_{i \in \{1, \dots, K\}} \|X - c_i\|^2.$$

The quantizer is *unconstrained* if the codewords $\{c_i\}$ are optimized freely (e.g., via Lloyd’s algorithm) without imposing any structural constraint such as lattice, product, or tree structure. The codebook is *optimal* if it minimizes the expected distortion $D(Q_K) = \mathbb{E}[\|X - c_{Q_K(X)}\|^2]$.

Definition 1.2 (Index Distribution). The *index distribution* of a quantizer Q_K for a source X with density f is the discrete distribution on $\{1, \dots, K\}$:

$$p_i = \mathbb{P}(Q_K(X) = i) = \int_{V_i} f(x) dx, \quad i = 1, \dots, K,$$

where $V_i = \{x \in \mathbb{R}^d : \|x - c_i\| \leq \|x - c_j\| \forall j\}$ is the Voronoi cell of codeword c_i . We write $p_{\text{index}} = (p_1, \dots, p_K)$.

Definition 1.3 (Entropy-Free vs. Entropy-Coded Rate). For a VQ with codebook size K and index distribution p_{index} :

- *Entropy-free rate* (fixed-length coding): $R_{\text{free}} = \lceil \log_2 K \rceil \approx \log_2 K$ bits/index.
- *Entropy-coded rate* (arithmetic/range coding): $R_{\text{coded}} = H(p_{\text{index}})$ bits/index, where $H(p) = -\sum_i p_i \log_2 p_i$ is the Shannon entropy.

The *rate penalty* is $\Delta R = R_{\text{free}} - R_{\text{coded}} = \log_2 K - H(p_{\text{index}})$.

Definition 1.4 (Maximum-Entropy Bound). The *maximum-entropy bound* for a K -ary distribution is $H_{\text{max}}(K) = \log_2 K$, achieved uniquely by the uniform distribution $U_K = (1/K, \dots, 1/K)$. For any distribution p on K symbols, $H(p) \leq \log_2 K$.

2 Maximum-Entropy Convergence of Unconstrained VQ

Lemma 2.1 (Equal-Mass Voronoi Cells). *Let X have a continuous density f on $\mathbb{R}^d$ with $f > 0$ a.e., and let $C_K^* = \{c_1^*, \dots, c_K^*\}$ be an optimal unconstrained codebook minimizing $D(Q_K)$. Then, as $K \rightarrow \infty$, the Voronoi cells V_i^* satisfy*

$$\mathbb{P}(X \in V_i^*) \rightarrow \frac{1}{K} \quad \text{for all } i = 1, \dots, K.$$

Proof. By the high-resolution quantization theory of Gersho & Gray (1992) and the equidistribution theorem for optimal quantizers (Gray & Neuhoff, 1998), an optimal quantizer for a smooth source density partitions the space into cells of approximately equal probability mass in the high-resolution regime.

More precisely, the point density $\lambda(x)$ of the optimal quantizer satisfies $\lambda(x) \propto f(x)^{d/(d+2)}$ (the Bennett–Gersho result), and the Voronoi cell volumes scale as $V_i \propto 1/(K \cdot \lambda(c_i))$. The probability mass of cell i is:

$$p_i = \int_{V_i} f(x) dx \approx f(c_i) \cdot \text{Vol}(V_i) \approx f(c_i) \cdot \frac{1}{K \cdot f(c_i)^{d/(d+2)}} = \frac{f(c_i)^{2/(d+2)}}{K}.$$

Since the normalization $\sum_i p_i = 1$ holds and the point density $\lambda(c_i) \propto f(c_i)^{d/(d+2)}$ is distributed to equalize the contribution of each cell to the total distortion, we obtain $p_i \rightarrow 1/K$ as $K \rightarrow \infty$.

Intuition: An optimal quantizer allocates codewords so that each cell contributes equally to the distortion. For a smooth density, this means each cell captures approximately $1/K$ of the probability mass. This is the *equidistribution property* of optimal quantizers. $\square$

Theorem 2.2 (Unconstrained VQ $\rightarrow$ Maximum Entropy). *Let Q_K be an optimal unconstrained vector quantizer with K codewords for a source X with continuous, positive density f on $\mathbb{R}^d$. Let $p_{\text{index}}^{(K)}$ denote the index distribution of Q_K . Then*

$$\lim_{K \rightarrow \infty} H(p_{\text{index}}^{(K)}) = \log_2 K,$$

i.e., the index distribution converges to the maximum-entropy (uniform) distribution.

Proof. By Lemma 2.1, the index probabilities satisfy $p_i^{(K)} \rightarrow 1/K$ for each i as $K \rightarrow \infty$.

The Shannon entropy of the index distribution is:

$$H(p_{\text{index}}^{(K)}) = -\sum_{i=1}^K p_i^{(K)} \log_2 p_i^{(K)}.$$

We measure the deviation from the maximum-entropy bound using the KL divergence from uniform:

$$\text{KL}\left(p_{\text{index}}^{(K)} \| U_K\right) = \sum_{i=1}^K p_i^{(K)} \log_2 \frac{p_i^{(K)}}{1/K} = \log_2 K - H\left(p_{\text{index}}^{(K)}\right).$$

By the equal-mass property (Lemma 2.1), for each i :

$$p_i^{(K)} = \frac{1}{K} \left(1 + \epsilon_i^{(K)}\right),$$

where $\epsilon_i^{(K)} \rightarrow 0$ as $K \rightarrow \infty$ and $\sum_i \epsilon_i^{(K)} = 0$ (since $\sum p_i = 1$).

Substituting:

$$\text{KL}\left(p_{\text{index}}^{(K)} \| U_K\right) = \sum_{i=1}^K \frac{1 + \epsilon_i^{(K)}}{K} \log_2 \left(1 + \epsilon_i^{(K)}\right).$$

Using the Taylor expansion $\log_2(1 + \epsilon) = \epsilon / \ln 2 - \epsilon^2 / (2 \ln 2) + O(\epsilon^3)$:

$$\text{KL} \approx \sum_{i=1}^K \frac{1 + \epsilon_i}{K} \left(\frac{\epsilon_i}{\ln 2} - \frac{\epsilon_i^2}{2 \ln 2} \right) = \frac{1}{K \ln 2} \sum_i \epsilon_i + \frac{1}{K \ln 2} \sum_i \epsilon_i^2 - \frac{1}{2K \ln 2} \sum_i \epsilon_i^2 + O(\epsilon^3).$$

Since $\sum_i \epsilon_i = 0$, the first term vanishes, and we get:

$$\text{KL}\left(p_{\text{index}}^{(K)} \| U_K\right) \approx \frac{1}{2K \ln 2} \sum_{i=1}^K \left(\epsilon_i^{(K)}\right)^2 = \frac{K}{2 \ln 2} \cdot \frac{1}{K} \sum_i \left(\epsilon_i^{(K)}\right)^2.$$

By the equidistribution theorem, the deviations $\epsilon_i^{(K)}$ satisfy $\frac{1}{K} \sum_i (\epsilon_i^{(K)})^2 = O(1/K^{2/d})$ (from the convergence rate of the quantizer point density, Graf & Luschgy, 2000), giving:

$$\text{KL}\left(p_{\text{index}}^{(K)} \| U_K\right) = O\left(K^{-2/d}\right) \rightarrow 0 \quad \text{as } K \rightarrow \infty.$$

Therefore:

$$H\left(p_{\text{index}}^{(K)}\right) = \log_2 K - \text{KL}\left(p_{\text{index}}^{(K)} \| U_K\right) \rightarrow \log_2 K \quad \text{as } K \rightarrow \infty. \quad \square$$

Remark 2.3. The convergence rate $O(K^{-2/d})$ means that for high-dimensional sources (large d), the convergence is slower. This is consistent with the curse of dimensionality in vector quantization. However, for the moderate dimensions typical in neural compression ($d = 8\text{--}256$), the convergence is practically fast enough that the entropy gap becomes negligible for codebook sizes $K \geq 2^{10}$.

3 Rate Penalty Bound

Theorem 3.1 (Rate Penalty Bound). *For a VQ with codebook size K and index distribution p_{index} , the rate penalty of entropy-free versus entropy-coded coding satisfies:*

$$\Delta R = R_{\text{free}} - R_{\text{coded}} = \log_2 K - H(p_{\text{index}}) \geq 0,$$

with equality if and only if p_{index} is uniform. Moreover, for an optimal unconstrained VQ (Theorem 2.2), $\Delta R \rightarrow 0$ as $K \rightarrow \infty$, with convergence rate $\Delta R = O(K^{-2/d})$.

Proof. Non-negativity. By the maximum-entropy property of the uniform distribution (Definition 1.4), $H(p) \leq \log_2 K$ for any distribution p on K symbols, with equality iff $p = U_K$. Therefore:

$$\Delta R = \log_2 K - H(p_{\text{index}}) \geq 0,$$

with equality iff $p_{\text{index}} = U_K$ (uniform distribution).

Asymptotic vanishing. By Theorem 2.2, $H(p_{\text{index}}^{(K)}) \rightarrow \log_2 K$ as $K \rightarrow \infty$ for unconstrained VQ. From the proof of Theorem 2.2, the rate of convergence is:

$$\Delta R = \text{KL}(p_{\text{index}} \| U_K) = O(K^{-2/d}).$$

Thus $\Delta R \rightarrow 0$ as $K \rightarrow \infty$.

Equivalence with KL divergence. Note that:

$$\Delta R = \log_2 K - H(p_{\text{index}}) = \sum_i p_i \log_2 \frac{p_i}{1/K} = \text{KL}(p_{\text{index}} \| U_K),$$

confirming that the rate penalty is exactly the KL divergence from the uniform distribution — a fundamental information-theoretic quantity measuring how far the index distribution is from maximum entropy. $\square$

Remark 3.2. The bound $\Delta R \leq \log_2 K$ is trivially achieved when $H(p_{\text{index}}) = 0$ (deterministic index, i.e., all source vectors map to a single codeword — a degenerate quantizer). The interesting regime is the *operational* one where the quantizer is useful (reasonable distortion), and the penalty is small due to near-uniformity.

4 Entropy-Free Optimality Conditions

Theorem 4.1 (Entropy-Free Optimality). *Entropy-free (fixed-length) coding achieves the same rate as entropy-coded coding for a VQ-based compression system if and only if the following three conditions hold:*

- (a) **Index uniformity:** $p_{\text{index}} \approx U_K$, i.e., $H(p_{\text{index}}) \approx \log_2 K$.
- (b) **Index independence:** the indices $\{Q_K(X_t)\}_{t=1}^n$ are mutually independent (no inter-latent correlation).
- (c) **Context irrelevance:** conditioning on any context variable C does not reduce the index entropy: $H(Q_K(X) | C) \approx H(Q_K(X))$.

Proof. We prove necessity and sufficiency for each condition.

(a) **Index uniformity.** If $p_{\text{index}} = U_K$, then $H(p_{\text{index}}) = \log_2 K$ and $\Delta R = 0$ by Theorem 3.1. Conversely, if $\Delta R = 0$, then $H(p_{\text{index}}) = \log_2 K$, which holds iff $p_{\text{index}} = U_K$.

(b) **Index independence.** For a sequence of indices $I_1, \dots, I_n$, the entropy-coded rate with joint context modeling is $H(I_1, \dots, I_n)/n$. If the indices are independent, $H(I_1, \dots, I_n) = \sum_t H(I_t)$, and the per-index rate is $\bar{H} = \frac{1}{n} \sum_t H(I_t)$. The entropy-free rate is $n \log_2 K / n = \log_2 K$ per index.

If indices are *correlated*, then $H(I_1, \dots, I_n) < \sum_t H(I_t)$ (by the chain rule and non-negativity of conditional mutual information). Entropy coding with a joint model achieves rate $H(I_1, \dots, I_n)/n < \bar{H}$, gaining over marginal entropy coding. Fixed-length coding cannot capture this correlation, so it pays an additional penalty:

$$\Delta R_{\text{corr}} = \bar{H} - \frac{H(I_1, \dots, I_n)}{n} = \frac{1}{n} \sum_t H(I_t) - \frac{1}{n} H(I_1, \dots, I_n) \geq 0.$$

Thus, independence is necessary for entropy-free optimality.

(c) Context irrelevance. If there exists a context C such that $H(I | C) < H(I)$, then conditioned entropy coding achieves rate $H(I | C) < H(I) \leq \log_2 K$, while entropy-free coding still uses $\log_2 K$ bits. The penalty is:

$$\Delta R_{\text{ctx}} = H(I) - H(I | C) = I(I; C) > 0.$$

Thus, context irrelevance ($I(I; C) \approx 0$) is necessary.

Sufficiency. If all three conditions hold, then:

- By (a): $H(p_{\text{index}}) = \log_2 K$, so $\Delta R = 0$.
- By (b): the joint entropy factorizes, so no inter-index gain.
- By (c): no context-based gain.

Therefore $R_{\text{free}} = R_{\text{coded}} = \log_2 K$ per index. $\square$

Remark 4.2 (Restoring Independence via Context-Conditioned Transforms). When condition (b) fails (correlated latents), EF-LIC proposes a *context-conditioned autoregressive transform* that removes inter-latent correlation without sequential entropy coding. The transform computes residuals:

$$R_t = Z_t - g(Z_{t-1}, Z_{t-2}, \dots, Z_{t-p}),$$

where g is a (learned) predictor. If g captures the predictable component, the residuals $\{R_t\}$ are approximately independent, restoring condition (b). Crucially, this transform is *parallelizable*: once g is learned, all residuals can be computed in parallel (unlike sequential entropy coding which requires processing symbols one at a time).

5 Speedup Characterization

Corollary 5.1 (Speedup from Entropy-Free Coding). *Removing entropy coding from the VQ compression pipeline yields approximately $3\times$ encoding speedup and $5\times$ decoding speedup, as reported in EF-LIC (Cao et al., 2026).*

Encoding. *The standard pipeline Transform $\rightarrow$ Quantize $\rightarrow$ Entropy Code has its bottleneck in the sequential entropy coding step. Replacing it with fixed-length index assignment eliminates the sequential dependency:*

$$T_{\text{encode}}^{\text{free}} \approx \frac{1}{3} T_{\text{encode}}^{\text{coded}},$$

yielding a $\sim 3\times$ speedup.

Decoding. *The standard pipeline Entropy Decode $\rightarrow$ Dequantize $\rightarrow$ Inverse Transform has its bottleneck in sequential entropy decoding. Fixed-length decoding reads all indices in parallel:*

$$T_{\text{decode}}^{\text{free}} \approx \frac{1}{5} T_{\text{decode}}^{\text{coded}},$$

yielding a $\sim 5\times$ speedup. The larger decode speedup reflects the fact that entropy decoding is a more severe bottleneck on the decode path (it must be perfectly sequential and bit-exact).

Rate cost. *By Theorem 3.1, the rate penalty is $\Delta R = O(K^{-2/d})$, which is negligible for practical codebook sizes.*

Proof. The speedup factors arise from the elimination of sequential dependencies:

Encoding. In the entropy-coded pipeline, encoding time is dominated by:

- Transform: $O(N)$ parallelizable (convolution/attention).
- Quantization: $O(N)$ parallelizable (nearest-neighbor lookup).
- Entropy coding: $O(N)$ *sequential* (arithmetic coding processes symbols one at a time, maintaining a running probability context).

The entropy coding step typically accounts for $\sim 2/3$ of the total encoding time (Cao et al., 2026, Table 3). Removing it reduces:

$$T_{\text{encode}}^{\text{free}} = T_{\text{transform}} + T_{\text{quant}} \approx \frac{1}{3} T_{\text{encode}}^{\text{coded}}.$$

Decoding. In the entropy-coded pipeline, decoding time is dominated by:

- Entropy decoding: $O(N)$ *sequential*, $\sim 4/5$ of total.
- Dequantization + inverse transform: $O(N)$ parallelizable.

The entropy decoding step accounts for $\sim 4/5$ of decoding time because arithmetic decoding is inherently sequential (each symbol depends on the accumulated bitstream state). Removing it:

$$T_{\text{decode}}^{\text{free}} = T_{\text{dequant}} + T_{\text{inverse}} \approx \frac{1}{5} T_{\text{decode}}^{\text{coded}}.$$

The rate cost of this speedup is $\Delta R = O(K^{-2/d})$ bits/index (Theorem 3.1), which is negligible for practical codebook sizes $K \geq 2^{10}$ and moderate dimensions $d \leq 64$. $\square$

6 Summary

Remark 6.1 (Summary of Results). The three theorems establish a complete characterization:

- **Theorem 1:** Unconstrained VQ naturally produces near-uniform index distributions ($H \rightarrow \log_2 K$), making entropy coding unnecessary in the asymptotic regime.
- **Theorem 2:** The rate penalty of skipping entropy coding is exactly the entropy gap $\Delta R = \log_2 K - H(p)$, which vanishes as $O(K^{-2/d})$.
- **Theorem 3:** Entropy-free coding is optimal iff indices are uniform, independent, and context-irrelevant. Correlations can be removed via parallelizable context-conditioned transforms.
- **Corollary:** The practical payoff is $3\times$ encode and $5\times$ decode speedup at negligible rate cost.

These results formalize the EF-LIC insight: unconstrained VQ + fixed-length coding achieves near-optimal compression with dramatically faster encoding and decoding by eliminating the sequential entropy coding bottleneck.

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